RechargeMatch: Product Requirements Document (v2.0)

Environmental Hacks, Bharat Builds Tour 2026 · Heat and Water track · October 2026 · Supersedes v1.0

1. Summary

One-line pitch: Delhi already knows where it floods. RechargeMatch shows where that water should go, how much the ground can realistically absorb, and which recharge structure to build first.

RechargeMatch is a ward-level planning tool for RWAs, NGOs and municipal engineers. It takes known and reported waterlogging hotspots, uses open terrain data to work out where the water comes from and where it can flow by gravity, and ranks nearby recharge sites that could capture it. For each match it reports two honest numbers: how much runoff a site can **capture** (flood relief) and how much of that can **infiltrate** (groundwater recharge). It also shows the **absorption gap**: how much runoff remains even after the best available sites are used.

What changed from v1.0 and why

v1.0 weakness	v2.0 fix
Matching was distance + capacity only, so it could recommend a site uphill of the flooded spot	Matching is terrain-aware: sites must be reachable by gravity from the hotspot's catchment
Real-time citizen diversion (sandbagging mid-flood) was the headline use case	Planning-first: pre-monsoon siting for ward planners. Citizen reports become evidence, not emergency instructions
Volume estimate used standing water (area × depth) and implied it all recharges	Separate captured vs recharged volumes, limited by site storage and infiltration over a design drain-down time
No water-quality or safety thinking	Safety gate: sewage-mixed water, silt, water-table depth and drain-down time are checked before any recommendation
Unsourced claims (for example, no competing software exists)	Claims are softened or flagged for verification (section 14)

v1.0 weakness	v2.0 fix
Six must-have features, four personas, a long list of 'or' tech choices	Four features, two users, one committed stack, a day-by-day cut list
Vanity metrics (70% 'accuracy' with no ground truth, latency)	Testable metrics: constraint correctness, expert back-test, transparency (section 10)
Demo relied on 'live monsoon reports' although the monsoon has ended	Storm-replay mode: a real past storm replayed over real hotspots

2. Problem

Indian cities such as Delhi suffer two linked failures each year. During intense monsoon rain, paved surfaces and clogged drains produce waterlogging and flooding. During the rest of the year, and especially in summer, groundwater levels fall and neighbourhoods depend on tankers and failing borewells. Much of the monsoon runoff is drained away rather than captured.

Existing tools handle other parts of the problem. Crowd-sourced flood reporting apps capture where flooding happens. Groundwater and reservoir dashboards monitor levels. Academic GIS studies identify dual-purpose flood and recharge zones. Physical measures such as recharge pits and permeable surfaces are known to work. What is hard to find, in the research we have done so far, is a citizen- and ward-facing tool that connects these pieces: from a specific hotspot to specific feasible sites, with transparent capacity estimates. We have not exhaustively surveyed the field, so the pitch says 'we did not find one' rather than 'none exists'.

Who feels it

- Residents and RWAs whose streets and underpasses flood repeatedly.
- Municipal and drainage engineers who must decide where to spend limited budget before the next monsoon.
- Groundwater-stressed areas where recharge has direct local value.

3. Product Thesis

1. **Planning first.** Waterlogging lasts hours, but hotspots recur every year. The highest-value decision is where to build capacity before the rain, not what to do mid-storm.
2. **Physics-aware.** Water follows terrain. A recommendation that ignores elevation and flow direction is not credible.

3. **Honest numbers.** Every figure is a range with its assumptions shown. A single site rarely solves a hotspot, and the product says so.

4. Users

User	Job to be done	What they get
Ward planner (RWA coordinator, municipal or DJB engineer, NGO)	Decide where to build the next recharge structures before monsoon and justify it	Ranked build list per ward, catchment map, capture and recharge estimates, absorption gap, export
Citizen reporter	Flag a waterlogged spot and see that it matters	Simple report form; confirmation that the report strengthened a hotspot; link to the ward plan

Citizens are a data source and an audience for the output. They are not asked to divert water during a flood.

5. Scope

5.1 MVP features (four)

F1. Hotspot layer and reporting

- Seed one demo ward with documented waterlogging hotspots (from publicly released lists and press reports, geocoded and cross-checked).
- Report form: map pin or auto-location, optional photo, depth category (ankle / knee / waist or above), duration category, notes.
- Each hotspot has a confidence level: seeded only, seeded + reports, or reports only.
- *Acceptance:* a report can be submitted in 4 fields or fewer; it appears on the map and updates the hotspot's confidence; photo EXIF location is stripped before storage.

F2. Terrain-aware matching

- Precomputed (offline) catchment and flow-path layers from an open DEM.
- Candidate sites from OpenStreetMap (parks, open ground, institutional campuses) plus a small hand-verified set in the demo ward.
- Hard filters, then weighted scoring, then top three ranked sites per hotspot with a plain-language reason for each (section 6).

- *Acceptance:* an automated test confirms that no recommended site is hydraulically uphill of its hotspot; every recommendation shows which filters and scores produced it.

F3. Capture and recharge estimator

- Per hotspot: runoff volume for a chosen storm. Per site: capturable volume and recharge volume, each with low / base / high values.
- Storm input: a user-chosen design storm, or **storm replay** of a real past rain event from open rainfall data.
- Ward-level outputs: total runoff, total capacity of recommended sites, absorption gap.
- *Acceptance:* every number on screen is labelled as an estimate, shows a range, and links to the assumptions used.

F4. Ward plan dashboard and export

- Map with hotspots, catchments, candidate sites and matched pairs; filters by confidence and by priority.
- Ranked 'build list': which site closes the most of the gap per unit of effort.
- CSV export of the build list and assumptions.
- *Acceptance:* a ward planner can go from open map to exported ranked list in under two minutes.

5.2 Stretch (only after F1-F4 are done and validated)

- Amazon Bedrock to generate plain-language explanations from the computed fields (never to produce numbers) and a photo-based severity hint for reports.
- Hindi and English interface.
- Rainfall forecast overlay for likely hotspots.
- Community follow-up photos for built structures.

5.3 Non-goals

- Real-time instructions to divert floodwater, or any automated control of gates or pumps.
- Integration with official SCADA or drainage network data.
- Hydrogeological or aquifer modelling.
- Training ML models on proprietary data.
- Offline sync, accounts and notifications in the MVP.

6. Matching Methodology

This is the core of the product, so it is specified in full.

6.1 Pipeline

1. **Catchment per hotspot.** From an open DEM (Copernicus GLO-30 or SRTM), compute flow direction and flow accumulation, then delineate the upstream catchment of each hotspot.
2. **Runoff coefficient.** Estimate land cover in the catchment from OpenStreetMap landuse and building footprints (paved, roof, green) and compute an area-weighted runoff coefficient.
3. **Candidate sites.** Extract open spaces and campuses within a search radius of each hotspot.
4. **Hard filters.** A site is excluded if it fails any of these:
   - It is not reachable by gravity: its elevation is not lower than the hotspot along a plausible flow path (with a small tolerance for DEM error).
   - It is itself flood-prone (inside the catchment's lowest zone or a recorded hotspot).
   - It is a water body, a road, or smaller than the minimum structure footprint.
5. **Scoring.** Remaining sites are scored on the weighted factors below.
6. **Safety gate.** Each match receives flags (section 6.3). Flags can downgrade a site to 'detention only' or exclude it.
7. **Output.** Top three sites per hotspot, each with score breakdown, suggested structure type and rough size, flags, and a confidence label.

6.2 Scoring factors (starting weights, tunable)

Factor	Weight	Notes
Flow-path distance from hotspot	25%	Along the flow path, not straight-line
Available area for storage	20%	Larger usable area, more capacity
Soil infiltration suitability	20%	From soil class (open global soil data); coarse
Depth to water table	15%	From coarse public groundwater-level data; shallow table lowers the score
Site flood exposure	10%	Penalise sites likely to be saturated in the same storm

Factor	Weight	Notes
Land-use and access feasibility	10%	Public park, institutional, or private (private lowers feasibility)

The weights are heuristics. They are exposed in the UI so a user can change them, and they are validated against the expert back-test (section 10).

6.3 Safety and quality gate

Floodwater in cities is often mixed with sewage, oil and silt. Recharging it untreated can degrade the groundwater the product is meant to protect. The engine therefore applies these rules:

- **Likely sewage-mixed or drain overflow** (hotspot beside a nala or sewer line): recommend detention or pre-treatment only; no direct recharge.
- **Silt and first flush:** every recharge recommendation includes a silt trap and first-flush diversion as part of the suggested structure.
- **Shallow water table:** flag when depth to water is small, since there is little unsaturated soil to filter and store water.
- **Sensitive neighbours:** flag proximity to landfills, industrial land or drinking-water sources, using what open data shows.
- **Drain-down time:** size storage so that it empties within a design time of about 48 hours, to avoid standing water and mosquito breeding.

These are screening rules, not engineering approval. Outputs say so (section 8).

6.4 Suggested structure types

The engine maps site type and soil to a suggestion only: recharge pit or trench with silt trap for open ground, edge trench for parks, detention basin where infiltration is poor, and rooftop-to-recharge connection for institutional campuses. Recharge wells are mentioned only for filtered water and need professional design.

7. Estimation Model

All values are reported as low / base / high.

Quantity	Formula	Inputs
Runoff volume (m^3)	$V_{runoff} = C \times P \times A$	C weighted runoff coefficient; P storm rainfall depth (m); A catchment area (m^2)
Site storage (m^3)	$V_{store} = L \times W \times D \times$ void factor	Suggested pit or trench size; void factor 1.0 for an open pit, lower if filled with media
Drain-limited volume (m^3)	$V_{drain} = f \times A_{floor} \times T$	f soil infiltration rate (m/h, with low / base / high by soil class); A_{floor} infiltrating area (m^2); T design drain-down time (h)
Design capacity per site (m^3)	$V_{design} = \min(V_{store}, V_{drain})$	Takes the smaller of the two
Captured per storm (m^3)	min(V_runoff delivered to the site, V_{design})	Flood relief
Recharge per storm (m^3)	Captured volume that infiltrates within T	Groundwater benefit
Annual recharge potential (m^3/yr)	Sum over qualifying storms in the rainfall record of the captured volume	Storms closer together than T are merged
Absorption gap (m^3)	V_{runoff} minus the sum of V_{design} for recommended sites	The planning headline

Runoff coefficients and infiltration rates are typical literature values and must be confirmed against CGWB and municipal guidance. A field infiltration test is required before construction.

Worked example (illustrative numbers)

- Catchment A = 20,000 m^2, storm P = 60 mm, C = 0.85, giving V_{runoff} = 0.85 × 0.060 × 20,000 = **1,020 m^3**.
- Candidate pit 10 m × 10 m × 3 m: V_{store} = **300 m^3**.
- Soil infiltration f = 20 mm/h, T = 48 h: V_{drain} = 0.020 × 100 × 48 = **96 m^3**.
- V_{design} = min(300, 96) = **96 m^3** (about 96,000 litres) captured per storm.
- Absorption gap = 1,020 − 96 = **924 m^3**. This one site absorbs about 9% of the storm, so the plan needs a mix of sites, larger infiltrating area, or upstream measures.

Showing the gap is deliberate. It makes the tool credible and tells planners where capacity is missing.

8. Safety, Legal and Ethics

- No real-time diversion advice. Guidance is addressed to planners and clearly states that any diversion or construction needs permission from the land owner or the municipal authority.
- Every screen with a recommendation carries: *Screening-level estimate. Needs site survey, infiltration test and professional hydrogeological review before construction.*
- A 30 m DEM cannot see storm drains, kerbs or underpass geometry. The tool uses terrain only for gross flow direction and gravity reachability, and says so.
- Photos and locations are personal data: strip EXIF, store minimal metadata, allow deletion, and do not show reporter identity.
- Reports can be wrong or malicious. Reports only raise a hotspot's confidence and never create a recommendation on their own; admin can hide a report.

9. Technical Architecture

One committed stack, using AWS where it helps.

Offline data pipeline (Python, run locally or in CI)

- DEM processing for flow direction and catchments with open-source tools (for example pysheds or WhiteboxTools).
- OpenStreetMap extraction (for example osmnx) for land cover and candidate sites.
- Outputs written as GeoJSON or Parquet to Amazon S3.

Runtime

- Frontend: React with MapLibre GL, deployed on AWS Amplify Hosting.
- API: Amazon API Gateway + AWS Lambda (Python, with shapely for geometry).
- Data: Amazon DynamoDB for hotspots, reports and matches; Amazon S3 for precomputed layers and photos.
- Maps: Amazon Location Service map tiles.
- No authentication in the MVP; rate limiting and request size limits at the API.

Why no spatial database: the demo ward has hundreds of candidate sites, so matching can run in memory in Lambda over precomputed layers. This avoids the geospatial limits of DynamoDB and the cost of an always-on database.

Optional: Amazon Bedrock for grounded explanations and photo triage (stretch only); AWS SAM CLI for local development.

Cost: confirm current AWS Free Tier terms before building and set a billing alarm on day 1.

Open data sources

Need	Source (verify access and licence)
Terrain	Copernicus GLO-30 DEM or SRTM
Land cover, sites	OpenStreetMap
Rainfall for storm replay	IMD records and NASA GPM IMERG
Soil class	Global open soil datasets (for example SoilGrids)
Groundwater depth	CGWB or India-WRIS (coarse, station-based)
Hotspot lists	Public PWD / MCD / Traffic Police lists and press reports

Core entities

Hotspot (location, catchment, confidence), Report (hotspot link, depth and duration category, photo reference, timestamp), Site (geometry, type, area, soil, flags), Match (hotspot, site, score breakdown, volumes low/base/high, flags), StormEvent (date, rainfall depth, source).

10. Success Metrics

Metric	Target	How it is measured
Constraint correctness	0 recommended sites hydraulically uphill of their hotspot	Automated test over all demo hotspots
Expert agreement	Top-3 list matches an expert's own top-3 for most hotspots	Back-test: one or two civil or water engineers rank sites for 8 to 10 hotspots independently; report overlap honestly, including misses
Estimate transparency	100% of displayed volumes show a range and assumptions	UI check

Metric	Target	How it is measured
Safety coverage	Every match carries its gate flags	Test
Data realism	At least 8 real hotspots, at least 30 candidate sites, at least 1 replayed real storm in the demo ward	Data checklist
Validation	At least 2 conversations with an RWA member, engineer or NGO; one quote or lesson in the demo	Interview notes
Ward query time	Under 3 seconds	Measured on the deployed URL

No claim of overall 'accuracy' is made. The honest claim is agreement with expert judgement on a small sample, plus guaranteed physical constraints.

11. Four-Day Build Plan

Day	Focus	Done when
1	Pick demo ward; seed hotspots and sites; run DEM and OSM pipeline; deploy skeleton on AWS; billing alarm	Map shows real hotspots, catchments and candidate sites on the deployed URL
2	Matching engine, safety gate, estimator, unit tests	Constraint test passes; worked example reproduces section 7
3	Report form, ward plan dashboard, export, storm replay	Full loop works end to end
4	Expert back-test, fixes, demo recording	Metrics measured; 3-minute video done

Feature freeze: end of Day 1. **Cut list, in order:** Bedrock features, photo upload, CSV export formatting, storm replay (fall back to a fixed design storm), editable weights.

12. Risks

Risk	Impact	Mitigation
30 m DEM too coarse for urban flow	High	Use only for gross flow direction; verify demo matches on the ground or with satellite imagery; label as screening-level
Incomplete or wrong site data	Medium	Hand-verify the demo ward; mark confidence; allow community site suggestions later
Infiltration and coefficients are assumptions	High	Show ranges; cite sources; require a field test; keep one transparent formula set
Perceived as telling people to divert floodwater	High	Planning-first framing; permission and safety notices on every recommendation
Contamination from recharging dirty water	High	Safety gate and mandatory silt trap and first flush; detention-only fallback
Little real report data after monsoon	Medium	Seeded real hotspots plus storm replay; do not claim live data
Scope creep	High	Four features, freeze after Day 1, explicit cut list
Weak adoption	Medium	Position as a planning aid for RWAs and engineers; show a concrete build list

13. Three-Minute Demo Script

1. **Hook (20s):** a flooded street in monsoon, then a dry borewell or tanker queue. 'Same neighbourhood, same year.'
2. **Insight (25s):** 'We know where it floods. We do not know where that water should go.' Show hotspots on the demo ward map.
3. **Walkthrough (95s):**
   - Pick one real hotspot; show its catchment and flow path.
   - Show the top three sites and why each ranked as it did, including one site rejected for being uphill and one flagged for sewage risk.

    - Replay a real past storm; show runoff, captured volume and recharge volume as ranges.
    - Show the absorption gap for the ward and the ranked build list; export it.
4. **Validation (20s):** one expert quote or the back-test result, including a miss.
5. **Tech (10s):** architecture diagram and deployed URL.
6. **Close (10s):** 'A ward that knows its gap can close it before the next monsoon.'

14. Assumptions and Items to Verify Before Submission

- Source and exact count of identified Delhi waterlogging-prone sites (v1.0 cited 169 without a source).
- Source for any groundwater over-exploitation statistics used on slides.
- Typical runoff coefficients and soil infiltration ranges, checked against CGWB and municipal guidance.
- Licences and access terms for the DEM, rainfall and soil datasets.
- Current hackathon rules for AWS usage and eligibility.
- Whether the demo ward's land parcels can be described as public, institutional or private.

End of document.